

Alexander L. Li

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EDUCATION

Massachusetts Institute of Technology (MIT)

Exp. May 2028

Candidate for B.S. in Artificial Intelligence (6-4) & Mathematics (18)

Cambridge, MA

- **Relevant Coursework:** Deep Learning, Natural Language Processing, Computer Vision, Algorithms, Web Design Lab, C & Assembly, Real Analysis, Discrete Math, Linear Algebra, Differential Equations, Probability & Statistics

TECHNICAL SKILLS

Languages: Python, C/C++, Golang, TypeScript, JavaScript, Java, SQL, HTML/CSS

Frameworks/Technologies: NumPy, PyTorch, TensorFlow, Git, Linux, Docker, React, Node.js, MongoDB, PostgreSQL, FastAPI, Flask, LangChain, LangGraph, SLURM, Apptainer/Singularity

Awards: USA Computing Olympiad Gold Division, 5th in Cryptography @ Science Olympiad Nationals

EXPERIENCE

Amazon Web Services (EC2/EBS Snapshots Team)

Jun 2026 – Aug 2026

Software Development Engineer (Internship / On-Site)

Boston, MA

- Raised data backup upload throughput by **2.41x** via new dynamic concurrency architecture & statistical methods
- Engineered lightweight C++ utility supporting latency tracking in a single pass, adoptable by other server modules
- Validated correctness & performance across 6 latency profiles via meticulous unit, integration, & A/B testing

Fetch.ai (Agentic AI Platform)

Jun 2025 – Sep 2025

Junior Software Engineer (Internship / Remote)

Cambridge, UK

- Shipped **5+** **agents** to production with **~10k user interactions**, facilitating dynamic service queries from LLMs
- Integrated multi-modal agent architecture by protocol-driven understanding & distributed agent orchestration
- Containerized agents with Docker, enabling edge-to-cloud communication by mailbox systems across agent nodes

Air Force Research Laboratory (AFRL)

May 2025 - Aug 2025

Deep Learning Researcher (Internship / On-site)

Wright-Patterson AFB, OH

- Trained a conditional diffusion model with PyTorch to reverse-engineer image generation parameters from Blender
- Designed multi-head attention U-Net conditioned on ResNet-encoded images to infer input parameter distributions
- Scaled training efficiency by **over 400%** via multi-node distributed data parallel (DDP) & batch scheduling
- Published as first author in SPIE Defense + Security Journal and presented work as a keynote speaker

PROJECTS

Deep Learning Chess Engine

Jan 2026

Python, PyTorch, Pygame, Modal, Reinforcement Learning, MCTS

- Engineered AlphaZero-style chess engine, coupling a dual-head ResNet CNN w/ Monte Carlo Tree Search (MCTS)
- Optimized parallel GPU evals via batched MCTS, leveraging UCB selection & LRU transposition tables
- Architected cloud-based self-play RL pipeline on Modal featuring supervised pre-training & auto checkpointing

Campus Marketplace Platform

Jul 2025

React.js, HTML, Tailwind CSS, Node.js, Express, FastAPI, PostgreSQL, Supabase, Docker, CLIP

- Built a scalable, mobile-first campus marketplace app with secure user auth & real-time item listings
- Implemented AI-powered category recommendations using OpenAI CLIP & FastAPI microservices
- Deployed in Dockerized containers with PostgreSQL & Supabase; architected for future cloud scaling

ChillDeck DJ (MIT Hackathon) - 2nd Place, Best UI Design

Jan 2025

React.js, HTML/CSS, TypeScript, RESTful APIs, Node.js, MongoDB, Figma

- Developed (as a team of 3) an interactive DJ platform allowing users to remix songs by isolating audio stems
- Engineered backend stem processing with Audioshake API & real-time audio waveforms via Wavesurfer.js
- Designed full user flow integrated with MongoDB: login, theming, uploads, & custom track playback